

Question 1.

```
select rider_type, count(*) as total_rides, sum(fare) as total_fare. from rides group by
rides having count(*) > 7 order by total_fare desc ;
```

Question 2

```
select station_id, count(*) as ride_count , ROUND AVG(duration_min,2) as
avg_duration
from rides group by station_id having AVG(duration_min)>30 order by
avg_duration desc ;
```

Question 3

```
select month(ride_data) as ride_month, count(*) as total_rides .,
round(sum(distance_km,1) as total_distance
from rides group by month (rides_date). having sum(distance_km,1) > 30 order by
ride_month ;
```

Question 4

```
select rider_type, max(duration_min) as longest_ride min(duration_min) as
shortes_ride
ROUND(avg(duration_min),2) as avg_ride
from ride group by rider_type having max(duration_min) >=50 order by longest_ride
desc;
```

Question 5

```
select station_id , rider_type, count(*) as ride_count , round(sum(distance_km,1) as
total_distance from rides group by station_id
Having (sum(distance_km,1)>10 order by station_id;
```

Question 7

```
select s.station_id , s.station_name , s.zone,
r.ride_id , r.rider_type , r.fare from station as s left join rides as r on s.station_id
```

= r.station\_id and month(r.ride\_date)=6 and year (r.ride\_date) = 2024 order by  
s.station\_id , r.ride\_id ;

#### Question 9

DDL , DML , DQL , DCL and TCL

DDL = DATA DEFINATION LANGUAGE IT IS USED TO [CREATE ,ALTER ,DROP,  
TRUNCATE] AND MODIFY DATABASE OBJECT

DML = DATA MANIPULATION LANGUAGE IT IS USED TO [ INSERT , UPDATE ,  
AND DELETE ]DATA

DQL = DATA QUERY LANGUAGE IT IS USED TO RETRIEVE DATA

DCL = DATA CONTROL LANGUAGE IT USED TO [GRANT AND REVOKE  
]PERMISSION

TCL = TRANSACTION CONTROL LANGUAGE IT IS USED TO MANAGE  
TRANSACTION [ COMMIT,ROLLBACK,SAVEPOINT]

#### Question 10

JOIN IS USED TO COMBINE ROWS FROM TWO OR MORE TABLES USING  
RELATED COLUMN

1 . INNER JOIN = RETURNS ONLY MATCHING DATA FROM TABLES

2 LEFT JOIN = RETURNS ALL RECORDS FROM LEFT TABLE AND MATCHING  
RECORDS FROM RIGHT TABLE

3 RIGHT JOIN = RETURNS ALL RECORDS FROM RIGHT TABLE AND MATCHING  
RECORDS FROM LEFT TABLE

4 FULL OUTER JOIN = RETURN ALL RECORDS OF BOTH TABLES NON

MATCHING VALUE WILL FILLED WITH NULL

5 CROSS JOIN

RETURNS EVERY POSSIBLE COMBINATION FROM BOTH TABLE (CARTESIAN PRODUCT ) generally we don't use this

Question 6

```
select station_id, count(distinct rider_type) as rider_type_count, count (*) as  
total_rides round(avg(fare),2) as avg_fare  
from rides group by station_id having count (distinct rider_type) =2 order by  
station_id;
```

Question 8